



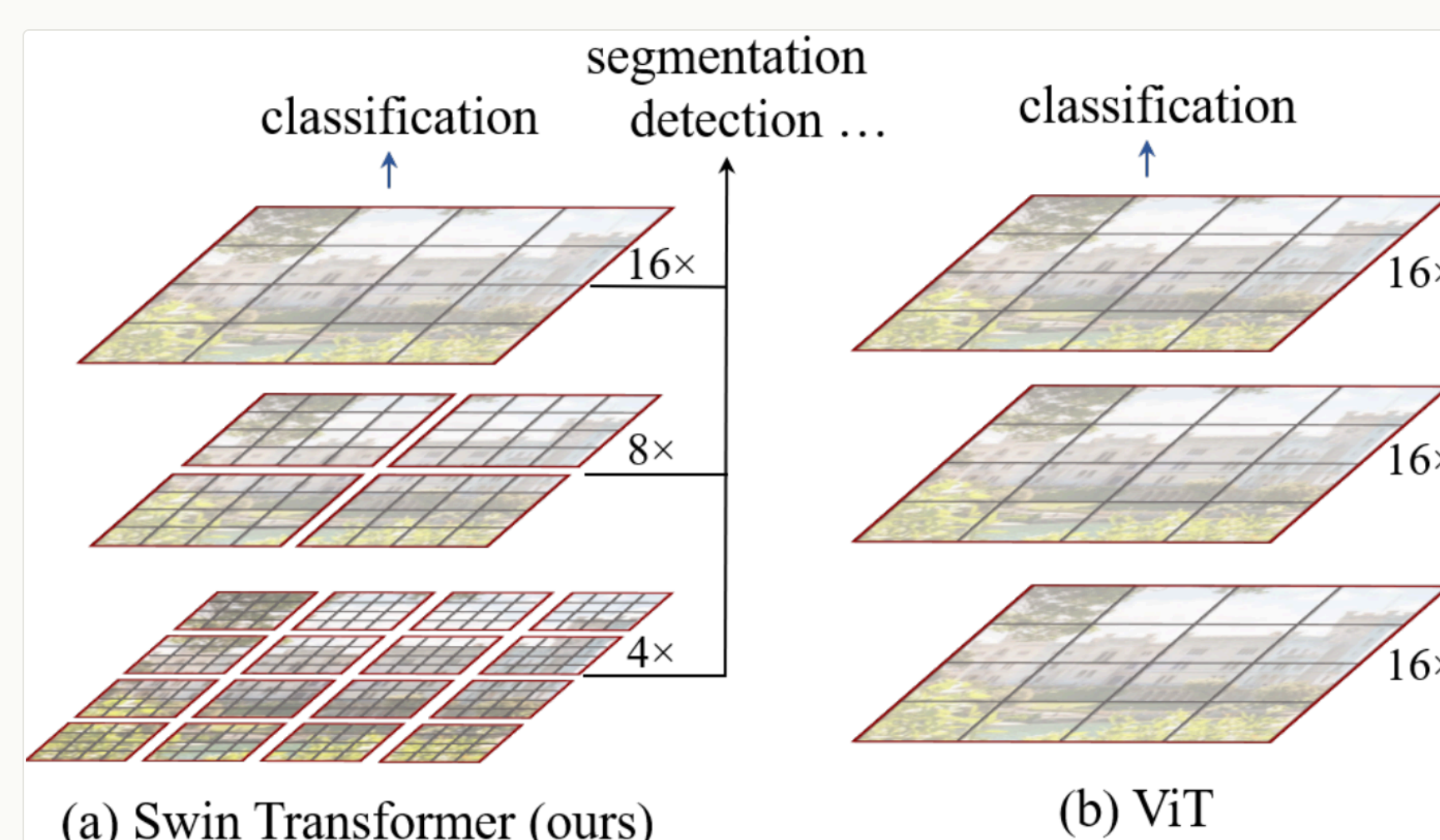
Problem

Standard vision Transformers like **ViT** compute **global self-attention** at one low resolution — **quadratic cost** in image size, and a single coarse feature map that detection and segmentation heads can't plug into.

ViT works for classification but isn't a general-purpose vision backbone — per-pixel tasks need both multi-scale features and tractable cost.

Motivation

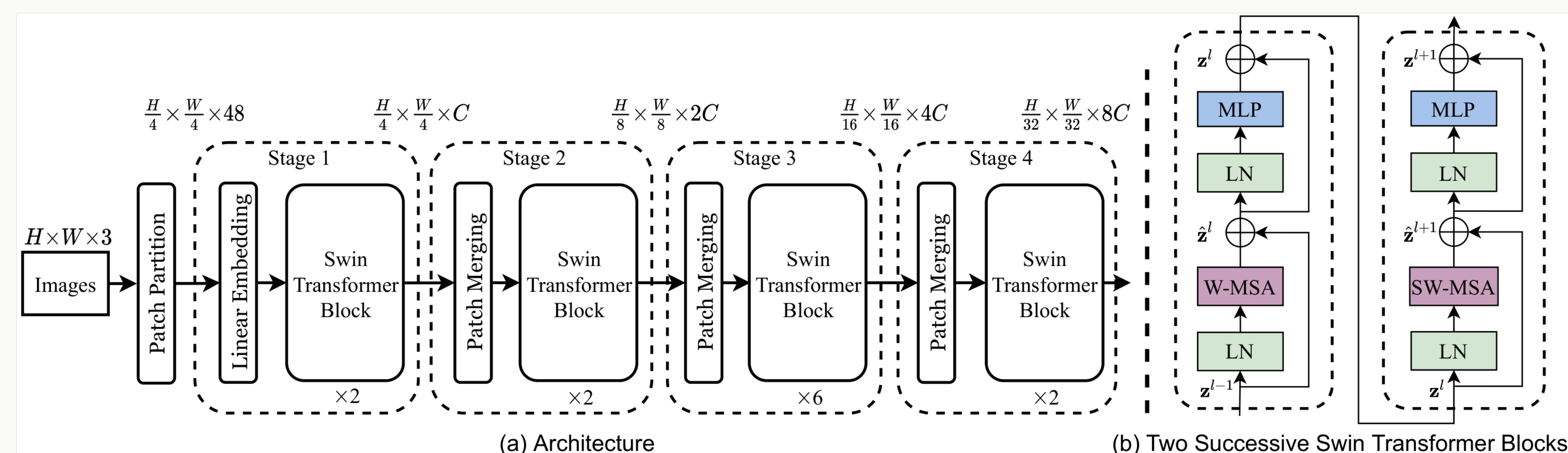
A general-purpose backbone needs **hierarchical multi-scale features** and **linear complexity**, without losing Transformer's long-range modeling. Sliding-window attention is accurate but slow in practice.



(a) Swin builds hierarchical feature maps via local-window self-attention (linear). (b) ViT yields one low-res map via global attention (quadratic).

Method

- 1 Split image into **4×4 patches**; self-attention **inside M×M windows** — **linear cost**. Next layer **shifts by [M/2]** to restore cross-window context.
- 2 **Patch-merging** → pyramid: **H/4, H/8, H/16, H/32**.



(a) Swin Transformer (Swin-T) architecture — four stages of patch-merging and Swin blocks. (b) Two successive Swin blocks alternating W-MSA (regular windows) and SW-MSA (shifted windows).

Datasets

ImageNet-1K **ImageNet-22K** **COCO 2017** **ADE20K**

- **Classification:** ImageNet-1K (1.28M imgs / 1000 classes) plus ImageNet-22K (14.2M / 22K) for large-scale pre-training.
- **Detection & instance seg:** COCO 2017 — 118K train, 5K val, 20K test-dev images.
- **Semantic seg:** ADE20K — 150 classes, dense per-pixel labels at high resolution.

Four Swin scales — **T** (29M params, 4.5G FLOPs), **S** (50M, 8.7G), **B** (88M, 15.4G), **L** (197M, 103.9G at 384²) — matched on compute to ResNet-50, ResNet-101, ViT-B and ViT-L respectively.

Key Results

BENCHMARK	PRIOR SOTA	SWIN-L
COCO — box AP	56.0	58.7 (+2.7)
COCO — mask AP	48.5	51.1 (+2.6)
ADE20K — mIoU	50.3	53.5 (+3.2)
ImageNet-1K — top-1	—	87.3%

One backbone, three new SOTAs — classification + detection + segmentation.

Swin-T beats **ResNet-50** by +3.4 to +4.2 box AP across four detection frameworks (Cascade R-CNN, ATSS, HTC, RepPoints v2) at matched latency — gains hold at every model scale.

Headline Numbers

87.3%

Top-1 · ImageNet-1K · Swin-L
w/ ImageNet-22K pretrain

58.7

COCO box AP

51.1

COCO mask AP

53.5

ADE20K mIoU

Ablation Study

- **Drop the shift** → −1.1% ImageNet, −2.8 box AP, −2.8 mIoU. Cross-window propagation is essential.
- **Relative position bias** beats both no-PE and absolute PE on all three benchmarks.

Cyclic-shift trick makes shifted-window SW-MSA as fast as W-MSA in practice.

Takeaway

Shifting non-overlapping local windows turns the Transformer into a hierarchical, linear-complexity vision backbone.

First Transformer to set SOTA on classification, detection, AND segmentation simultaneously — now a **drop-in ResNet replacement** for dense prediction.